Elementary Analysis

Chapter 4, Sections 23–27

All 56 exercises, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Sections 23–27, 56 exercises. Short descriptions identify the problems; consult the linked second-edition text for the original statements.

Contents

23 Power series **4**

23.1 Exercise 23.1: Eight radii and their endpoint tests 4
23.2 Exercise 23.2: Four more series, including missing powers 5
23.3 Exercise 23.3: The endpoint test for the sparse series in Example 6 6
23.4 Exercise 23.4: Roots and ratios with oscillating coefficients 6
23.5 Exercise 23.5: Two coefficient conditions forcing radius at most one 7
23.6 Exercise 23.6: Endpoint asymmetry and nonnegative coefficients 7
23.7 Exercise 23.7: Powers of cosine and pointwise limits 7
23.8 Exercise 23.8: Convergence of functions does not imply convergence of derivatives . 8
23.9 Exercise 23.9: A vanishing pointwise limit with nonvanishing integrals 8

24 Uniform convergence **9**

24.1 Exercise 24.1: One bound for every oscillation . . . 9
24.2 Exercise 24.2: A small slope on bounded and unbounded domains . . . 9
24.3 Exercise 24.3: A transition concentrated near one . . . 10
24.4 Exercise 24.4: The complementary transition . . . 10
24.5 Exercise 24.5: A denominator containing both the index and a power . . . 11
24.6 Exercise 24.6: Shifted squares . . . 11
24.7 Exercise 24.7: Subtracting a power . . . 11
24.8 Exercise 24.8: Partial geometric sums at their divergent endpoint . . . 12
24.9 Exercise 24.9: A moving peak whose integral vanishes . . . 12
24.10 Exercise 24.10: Adding uniform limits, and the danger with products . . . 13
24.11 Exercise 24.11: The product counterexample in full . . . 13
24.12 Exercise 24.12: The supremum-error characterization . . . 14
24.13 Exercise 24.13: Uniform limits of uniformly continuous functions . . . 14
24.14 Exercise 24.14: A narrow peak near zero . . . 14
24.15 Exercise 24.15: A boundary layer with limiting height one . . . 15
24.16 Exercise 24.16: A boundary layer with an unbounded limiting function . . . 15
24.17 Exercise 24.17: Passing to the limit while the input also moves . . . 16

25 More on uniform convergence 17

25.1 Exercise 25.1: Bounding an integral by the integral of an absolute value . . . 17
25.2 Exercise 25.2: Powers divided by their indices . . . 17
25.3 Exercise 25.3: A rational trigonometric expression and its integral . . . 17
25.4 Exercise 25.4: Uniform convergence implies the uniform Cauchy property . . . 18
25.5 Exercise 25.5: A uniform limit of bounded functions is bounded . . . 18
25.6 Exercise 25.6: Absolute summability gives continuity at the endpoints too . . . 18
25.7 Exercise 25.7: A uniformly convergent trigonometric series . . . 19
25.8 Exercise 25.8: A power series uniform on its entire convergence interval . . . 19
25.9 Exercise 25.9: Geometric series on a smaller and a maximal open interval . . . 20
25.10 Exercise 25.10: A non-geometric series compared with a geometric one . . . 20
25.11 Exercise 25.11: Four increasingly compressed triangular waves . . . 21
25.12 Exercise 25.12: Integrating a uniformly convergent series . . . 21
25.13 Exercise 25.13: Adding two uniformly convergent series . . . 22
25.14 Exercise 25.14: Multiplying a series by one bounded function . . . 22
25.15 Exercise 25.15: Monotone convergence on a compact interval: Dini's theorem . . . 23

26 Differentiation and integration of power series **24**

26.1 Exercise 26.1: Term-by-term integration at a positive input 24

26.2 Exercise 26.2: A differentiated geometric series and three numerical sums 24

26.3 Exercise 26.3: One more derivative gives squared indices 25

26.4 Exercise 26.4: Integrating the Gaussian power series 26

26.5 Exercise 26.5: A function equal to its own derivative, without naming it 26

26.6 Exercise 26.6: A trigonometric identity derived without trigonometry 27

26.7 Exercise 26.7: Why absolute value has no power-series representation at zero 27

26.8 Exercise 26.8: The arctangent series and its endpoints 28

27 Weierstrass's approximation theorem **30**

27.1 Exercise 27.1: Transferring approximation from $[0, 1]$ to $[a, b]$ 30

27.2 Exercise 27.2: One sequence that works on every bounded subset of $\mathbb{R}$ 30

27.3 Exercise 27.3: Why the entire real line is too large 31

27.4 Exercise 27.4: Correcting the two endpoint values 31

27.5 Exercise 27.5: Bernstein polynomials for the first two powers 32

27.6 Exercise 27.6: A necessary condition for uniform Bernstein approximation 33

27.7 Exercise 27.7: A Bernstein bound from nonnegative weights 33

23 Power series

For a power series $\sum_{n\geq 0} a_n x^n$, write $\beta = \limsup_{n\to\infty} |a_n|^{1/n}$. The radius of convergence is $R = 1/\beta$, with $1/0 = \infty$ and $1/\infty = 0$. The series converges absolutely for $|x| < R$ and diverges for $|x| > R$. *Neither endpoint is decided by the radius calculation*: substitute $x = R$ and $x = -R$ into the original series and test them separately. Finite initial terms do not affect convergence. Thus all series containing $1/n$ start at $n = 1$, and harmless choices of starting index do not change the answers. We use $n^{c/n} \to 1$ for fixed c, which follows by taking logarithms: $\log(n^{c/n}) = c \log n/n \to 0$.

23.1 Exercise 23.1: Eight radii and their endpoint tests

What we must show. Find the exact convergence interval in each case.

(a) The series $\sum n^2 x^n$.

Here $|a_n|^{1/n} = n^{2/n} \to 1$, so $R = 1$. At $x = 1$ the terms are n^2, and at $x = -1$ they are $(-1)^n n^2$. In both cases the terms fail to tend to zero, a necessary condition for series convergence. Thus the exact interval is $\boxed{(-1, 1)}$.

(b) The series $\sum (x/n)^n$.

The coefficient of x^n is $1/n^n$, whose nth root is $1/n \to 0$. Thus $R = \infty$. To see this directly, for fixed real x and sufficiently large n we have $|x|/n \leq 1/2$, and hence $|(x/n)^n| \leq 2^{-n}$. The geometric comparison proves absolute convergence. The exact interval is $\boxed{\mathbb{R}}$.

(c) The series $\sum (2^n/n^2)x^n$.

The coefficient root $2/n^{2/n}$ tends to 2, so $R = 1/2$. At $x = 1/2$ the series becomes $\sum 1/n^2$, which converges. At $x = -1/2$ it becomes $\sum (-1)^n/n^2$, which converges absolutely by the same p-series. The exact interval is $\boxed{[-1/2, 1/2]}$.

(d) The series $\sum (n^3/3^n)x^n$.

The coefficient root $n^{3/n}/3$ tends to $1/3$, giving $R = 3$. At $x = 3$ the terms are n^3; at $x = -3$ they are $(-1)^n n^3$. Neither tends to zero. The exact interval is $\boxed{(-3, 3)}$.

(e) The series $\sum (2^n/n!)x^n$.

For $x \neq 0$ the ratio of successive absolute terms is

$$\frac{2^{n+1}|x|^{n+1}/(n+1)!}{2^n|x|^n/n!} = \frac{2|x|}{n+1} \longrightarrow 0 < 1.$$

The ratio test proves absolute convergence for every such x. At $x = 0$ convergence is immediate. Thus $R = \infty$ and the interval is $\boxed{\mathbb{R}}$.

(f) The series $\sum x^n/((n+1)^2 2^n)$.

Its coefficient root is $1/(2(n+1)^{2/n}) \to 1/2$, so $R = 2$. At $x = 2$ it is $\sum 1/(n+1)^2$; at $x = -2$ it is $\sum (-1)^n/(n+1)^2$. Both converge absolutely. The interval is $\boxed{[-2, 2]}$.

(g) The series $\sum (3^n/(n4^n))x^n$.

The coefficient root $(3/4)n^{-1/n}$ tends to $3/4$, so $R = 4/3$. At $x = 4/3$ we obtain the divergent harmonic series $\sum 1/n$. At $x = -4/3$ we obtain $\sum(-1)^n/n$, which converges by the alternating-series test: $1/n$ decreases to zero. It is conditionally convergent because its absolute-value series is harmonic. The interval is $\boxed{[-4/3, 4/3)}$.

(h) The series $\sum((-1)^n/(n^2 4^n))x^n$.

Taking absolute values before the root removes the alternating sign: $|a_n|^{1/n} = 1/(4n^{2/n}) \to 1/4$. Hence $R = 4$. At $x = 4$ the series is $\sum(-1)^n/n^2$, and at $x = -4$ it is $\sum 1/n^2$, since the two factors $(-1)^n$ multiply to 1. Both converge absolutely, giving $\boxed{[-4, 4]}$.

Conclusion. The radii in order are $1, \infty, 1/2, 3, \infty, 2, 4/3, 4$. The separate endpoint tests determine the displayed intervals.

23.2 Exercise 23.2: Four more series, including missing powers

(a) The series $\sum \sqrt{n}\, x^n$.

The coefficient root is $n^{1/(2n)} \to 1$, giving $R = 1$. At $x = 1$ and $x = -1$, the term magnitudes are $\sqrt{n}$, which do not tend to zero. Thus the exact interval is $\boxed{(-1, 1)}$.

(b) The series $\sum x^n/(n\sqrt{n})$.

The denominator here is $n\sqrt{n} = n^{3/2}$, not an nth root. We have $|a_n|^{1/n} = n^{-3/(2n)} \to 1$, so $R = 1$. At either endpoint the absolute series is $\sum n^{-3/2}$, a convergent p-series because $3/2 > 1$. Thus the exact interval is $\boxed{[-1, 1]}$.

(c) The sparse series $\sum x^{n!}$.

For $|x| < 1$, since $n! \geq n$, $|x|^{n!} \leq |x|^n$. The right side has a convergent geometric sum, so absolute convergence follows. At $x = 1$ every term is 1. At $x = -1$, all terms with $n \geq 2$ are 1 because $n!$ is even. Both endpoint series diverge. For $|x| > 1$, the term magnitudes $|x|^{n!}$ do not approach zero. Therefore the radius is $R = 1$ and the exact interval is $\boxed{(-1, 1)}$.

(d) The series $\sum(3^n/\sqrt{n})x^{2n+1}$.

The exponent is $2n + 1$, so treating $3^n/\sqrt{n}$ as the coefficient of x^n would give the wrong radius. Instead write

$$\sum_{n\geq 1} \frac{3^n}{\sqrt{n}} x^{2n+1} = x \sum_{n\geq 1} \frac{(3x^2)^n}{\sqrt{n}}.$$

For $x \neq 0$, the absolute-term ratio tends to $3x^2$. Thus the series converges absolutely when $3x^2 < 1$ and diverges when $3x^2 > 1$. At $x = 0$ all terms vanish. At $x = 1/\sqrt{3}$, its terms equal $1/(\sqrt{3}\sqrt{n})$; at $x = -1/\sqrt{3}$, they equal $-1/(\sqrt{3}\sqrt{n})$. Each endpoint gives a constant multiple of the divergent p-series $\sum n^{-1/2}$. Thus

$$\boxed{R = 1/\sqrt{3}, \qquad \text{interval } (-1/\sqrt{3}, 1/\sqrt{3}).}$$

Conclusion. Missing powers must be respected. Testing the actual terms, rather than a misindexed coefficient sequence, avoids that error.

23.3 Exercise 23.3: The endpoint test for the sparse series in Example 6

What we must show. Determine the exact interval for $\sum_{n\geq 0} 2^{-n}x^{3n}$.

Recognize a geometric series in a new quantity.

Since $2^{-n}x^{3n} = (x^3/2)^n$, this is geometric with ratio $r = x^3/2$. It converges exactly when $|r| < 1$, equivalently $|x| < 2^{1/3}$. At $x = 2^{1/3}$ the terms are 1; at $x = -2^{1/3}$ they are $(-1)^n$. Neither term sequence tends to zero.

Conclusion. The exact interval is $(-2^{1/3}, 2^{1/3})$. The radius is $2^{1/3}$, not 2, because the powers are x^{3n}.

23.4 Exercise 23.4: Roots and ratios with oscillating coefficients

What we must show. Analyze $a_n = [(4 + 2(-1)^n)/5]^n$.

Separate even and odd indices before calculating.

For even n, $a_n = (6/5)^n$; for odd n, $a_n = (2/5)^n$. The term $a_0 = 1$ does not affect any limiting or convergence question.

(a) Compute the four requested extended limits.

The sequence $a_n^{1/n}$ alternates between $6/5$ and $2/5$, so

$$\limsup a_n^{1/n} = 6/5, \qquad \liminf a_n^{1/n} = 2/5.$$

For even n, the next index is odd and

$$\frac{a_{n+1}}{a_n} = \frac{(2/5)^{n+1}}{(6/5)^n} = \frac{2}{5}\,3^{-n} \longrightarrow 0.$$

For odd n, the next index is even and

$$\frac{a_{n+1}}{a_n} = \frac{(6/5)^{n+1}}{(2/5)^n} = \frac{6}{5}\,3^{n} \longrightarrow +\infty.$$

All coefficients are positive, so absolute values change nothing. Thus the ratio limsup is $+\infty$ and the ratio liminf is 0.

(b) Apply the necessary term test to the two numerical series.

Along even indices, $a_{2k} = (6/5)^{2k} \to +\infty$. Hence a_n does not tend to zero and $\sum a_n$ diverges. Also $(-1)^{2k}a_{2k} = a_{2k}$, so the signed series $\sum(-1)^n a_n$ diverges for the same reason. Alternating signs cannot make a series converge unless its terms tend to zero.

(c) Find the power-series interval.

The root limsup gives $R = 1/(6/5) = 5/6$. At either $x = 5/6$ or $x = -5/6$, the even terms satisfy

$$a_{2k}x^{2k} = (6/5)^{2k}(5/6)^{2k} = 1.$$

Neither endpoint series has terms tending to zero. Therefore

$$\boxed{R = 5/6, \qquad \text{interval } (-5/6, 5/6).}$$

Conclusion. The ordinary ratio limit does not exist. Its extreme subsequences illustrate why the root limsup is the general formula.

23.5 Exercise 23.5: Two coefficient conditions forcing radius at most one

(a) Infinitely many nonzero integer coefficients.

A nonzero integer has absolute value at least 1. There are therefore arbitrarily large indices n with $|a_n|^{1/n} \geq 1$. Consequently $\limsup |a_n|^{1/n} \geq 1$, and the reciprocal-radius formula yields $R \leq 1$. If that limsup is infinite, the radius is zero, which also satisfies the conclusion.

(b) A positive limsup of coefficient magnitudes.

If $\limsup |a_n| > 0$ (including $+\infty$), choose $c > 0$ below this limsup. Infinitely many indices $n_j \to \infty$ satisfy $|a_{n_j}| > c$. At those indices $|a_{n_j}|^{1/n_j} \geq c^{1/n_j}$, and $c^{1/n_j} \to 1$. Thus $\limsup_n |a_n|^{1/n} \geq 1$ and again $R \leq 1$.

Conclusion. The coefficients need not stay large at every index. A single infinite subsequence bounded away from zero already suffices.

23.6 Exercise 23.6: Endpoint asymmetry and nonnegative coefficients

(a) Convergence at the positive endpoint implies convergence at the negative one.

Suppose $a_n \geq 0$ and the finite radius is $R \geq 0$. If $R > 0$ and $\sum a_n R^n$ converges, then

$$\sum |a_n(-R)^n| = \sum a_n R^n < \infty.$$

Hence the series at $-R$ converges absolutely. If $R = 0$, the two endpoints are the same point and the assertion is immediate. The sign hypothesis is doing the work: the positive-endpoint sum is already a sum of absolute values.

(b) Construct the opposite endpoint pattern.

Take

$$\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} x^n.$$

Its coefficient-root limsup is 1, so its radius is 1. At $x = 1$ the alternating harmonic series converges. At $x = -1$ the series is $-\sum 1/n$, which diverges. Thus the exact interval is $\boxed{(-1, 1]}$. These coefficients do not satisfy the nonnegative hypothesis of part (a), so there is no contradiction.

Conclusion. Endpoint behavior depends on signs as well as magnitudes.

23.7 Exercise 23.7: Powers of cosine and pointwise limits

What we must show. Analyze $f_n(x) = (\cos x)^n$ for each fixed real x.

(a) Points that are not multiples of π.

At such a point $-1 < \cos x < 1$. Therefore $|\cos x|^n \to 0$, and $|f_n(x)| \to 0$ implies $f_n(x) \to 0$.

(b) Even multiples of π.

If $x = 2k\pi$ for an integer k, then $\cos x = 1$, so $f_n(x) = 1$ for every n and the limit is 1.

(c) Odd multiples of π.

If $x = (2k+1)\pi$, then $\cos x = -1$, so $f_n(x) = (-1)^n$. The even subsequence is constantly 1 and the odd subsequence is constantly -1. Since they have different limits, the full limit does not exist.

Conclusion. The three cases exhaust the real line. Continuity of every f_n does not even guarantee pointwise convergence at every point.

23.8 Exercise 23.8: Convergence of functions does not imply convergence of derivatives

What we must show. Compare $f_n(x) = \sin(nx)/n$ with its derivative.

(a) Bound the function values independently of x.

For all real x,

$$|f_n(x)| \le 1/n \longrightarrow 0.$$

The squeeze theorem gives $f_n(x) \to 0$ for every x. In fact this estimate proves uniform convergence, because the upper bound is the same for all x.

(b) Differentiate and test a specific point.

The chain rule gives $f_n'(x) = \cos(nx)$. At $x = \pi$, $f_n'(\pi) = (-1)^n$, so the derivatives have no pointwise limit there. Even at $x = 0$ they are always 1, whereas the derivative of the limit function $f \equiv 0$ is 0.

Conclusion. The operation of taking a derivative cannot be passed through a function limit without additional hypotheses. The oscillations get small in height while their slopes do not.

23.9 Exercise 23.9: A vanishing pointwise limit with nonvanishing integrals

What we must show. Analyze $f_n(x) = nx^n$ on $[0, 1]$.

(a) Fix a point strictly below one.

At $x = 0$ all values are zero. If $0 < x < 1$, set $u_n = nx^n > 0$. Its successive ratio satisfies $u_{n+1}/u_n = (1 + 1/n)x \to x < 1$. Choose q with $x < q < 1$. For some N, $u_{n+1} \le qu_n$ for all $n \ge N$. Repeatedly applying this inequality gives $u_{N+j} \le u_N q^j \to 0$. Thus $f_n(x) \to 0$ for every $x \in [0, 1)$. At $x = 1$, however, $f_n(1) = n \to +\infty$, so no finite limit exists there.

(b) Evaluate each integral before taking the limit.

For fixed n, the ordinary power rule for integration yields

$$\int_0^1 f_n(x)\,dx = n\left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{n}{n+1} = 1 - \frac{1}{n+1} \longrightarrow 1.$$

Conclusion. The functions approach zero at every point short of the right endpoint, but their integrals approach 1. Mass concentrates closer and closer to 1; pointwise information alone does not justify interchanging limit and integral.

24 Uniform convergence

Before starting. Pointwise convergence means fixing one input x first and then letting n grow. Uniform convergence means finding one index that works for *all* inputs at once. We shall repeatedly estimate the largest error

$$E_n = \sup_{x \in S} |f_n(x) - f(x)|.$$

Uniform convergence is equivalent to $E_n \to 0$; Exercise 24.12 proves this. To disprove uniform convergence, it is enough to find a positive number ε_0 and inputs $x_n \in S$, for arbitrarily large n, with $|f_n(x_n) - f(x_n)| \geq \varepsilon_0$. The input is allowed to change with n. That is precisely the obstacle uniform convergence must overcome.

24.1 Exercise 24.1: One bound for every oscillation

What we must show. Prove uniform convergence to zero for $f_n(x) = (1 + 2\cos^2(nx))/\sqrt{n}$ on $\mathbb{R}$.

Plan. We need not follow the oscillations: the cosine is always between -1 and 1.

Bound the numerator independently of the input.

Since $0 \leq \cos^2(nx) \leq 1$, we have

$$|f_n(x) - 0| \leq \frac{3}{\sqrt{n}} \qquad (x \in \mathbb{R}).$$

Given $\varepsilon > 0$, choose an integer $N > (3/\varepsilon)^2$. For every $n > N$ and every real x, the last bound is smaller than ε. The choice of N depends only on ε.

Conclusion. $f_n \to 0$ uniformly on $\mathbb{R}$.

24.2 Exercise 24.2: A small slope on bounded and unbounded domains

What we must show. Analyze $f_n(x) = x/n$ on $[0, \infty)$.

(a) Fix the input before passing to the limit.

For fixed $x \geq 0$, the numerator is constant and $n \to \infty$, so $f_n(x) \to 0$. Thus the pointwise limit is $f(x) = 0$.

(b) Control all inputs in $[0, 1]$.

There $|f_n(x)| \leq 1/n$. Given $\varepsilon > 0$, any integer $N > 1/\varepsilon$ works for all $x \in [0, 1]$ and $n > N$.

(c) Let the input escape along with the index.

On $[0, \infty)$ take $x_n = n$. Then $f_n(x_n) = 1$, so the error never falls below $1/2$. This contradicts the requirement of uniform convergence.

Conclusion. The limit is zero everywhere; convergence is uniform on $[0, 1]$ but not on $[0, \infty)$.

24.3 Exercise 24.3: A transition concentrated near one

What we must show. Analyze $f_n(x) = 1/(1+x^n)$ for $x \geq 0$.

(a) Separate the three possible behaviors of a power.

If $0 \leq x < 1$, then $x^n \to 0$, giving limit 1. At $x = 1$ every term is $1/2$. If $x > 1$, then $x^n \to \infty$, giving limit 0. Thus

$$f(x) = \begin{cases} 1 & 0 \leq x < 1, \\ 1/2 & x = 1, \\ 0 & x > 1. \end{cases}$$

(b) Test uniform convergence on $[0,1]$.

Each f_n is continuous there because its denominator is positive. The limit is not continuous at 1: its values from the left are 1, whereas $f(1) = 1/2$. The uniform-limit theorem for continuous functions therefore excludes uniform convergence on $[0,1]$.

(c) Use restriction to a smaller domain.

Uniform convergence on $[0,\infty)$ would imply uniform convergence on its subset $[0,1]$, with exactly the same error bound. Part (b) rules that out.

Conclusion. The piecewise function above is the pointwise limit, and neither of the two requested uniform-convergence assertions holds.

24.4 Exercise 24.4: The complementary transition

What we must show. Analyze $f_n(x) = x^n/(1+x^n)$ on $[0,\infty)$.

(a) Use an exact identity.

We have $f_n(x) = 1 - 1/(1+x^n)$. The three cases of the preceding exercise therefore give

$$f(x) = \begin{cases} 0 & 0 \leq x < 1, \\ 1/2 & x = 1, \\ 1 & x > 1. \end{cases}$$

(b) Notice the endpoint jump.

The f_n are continuous on $[0,1]$, but f is not continuous at 1. Consequently convergence cannot be uniform there, by Theorem 24.3.

(c) Restrict any proposed larger-domain convergence.

Uniform convergence on $[0,\infty)$ would restrict to uniform convergence on $[0,1]$, contradicting part (b).

Conclusion. The displayed f is the limit; convergence is uniform on neither $[0,1]$ nor $[0,\infty)$.

24.5 Exercise 24.5: A denominator containing both the index and a power

What we must show. Analyze $f_n(x) = x^n/(n + x^n)$ for $x \geq 0$.

(a) Find the pointwise limit.

For $0 \leq x \leq 1$, we have $0 \leq f_n(x) \leq 1/n$, so the limit is zero. For fixed $x > 1$, divide numerator and denominator by x^n:

$$f_n(x) = \frac{1}{1 + n/x^n} \longrightarrow 1.$$

Here $n/x^n \to 0$ because exponential growth dominates the linear factor; for example, the ratio of successive terms tends to $1/x < 1$. Thus $f(x) = 0$ on $[0, 1]$ and $f(x) = 1$ on $(1, \infty)$.

(b) Obtain a common bound on $[0, 1]$.

The estimate $|f_n(x) - f(x)| \leq 1/n$ works simultaneously for every input in that interval. Hence convergence is uniform there.

(c) Use the jump at an interior point of the larger domain.

Every f_n is continuous on $[0, \infty)$, but the pointwise limit jumps at 1. The uniform-limit theorem excludes uniform convergence on the larger domain.

Conclusion. Uniform convergence holds on $[0, 1]$, and fails on $[0, \infty)$.

24.6 Exercise 24.6: Shifted squares

What we must show. Find the limit of $f_n(x) = (x - 1/n)^2$ on $[0, 1]$ and decide uniform convergence.

(a) Expand to identify the limit.

For fixed x, $f_n(x) = x^2 - 2x/n + 1/n^2 \to x^2$. Thus $f(x) = x^2$.

(b) Bound the entire error, rather than just each separate input.

For $0 \leq x \leq 1$,

$$|f_n(x) - x^2| = \left| -\frac{2x}{n} + \frac{1}{n^2} \right| \leq \frac{2}{n} + \frac{1}{n^2} \leq \frac{3}{n}.$$

Given $\varepsilon > 0$, choose $N > 3/\varepsilon$. This bound is then less than ε for every $x \in [0, 1]$ and $n > N$.

Conclusion. $f_n \to (x \mapsto x^2)$ uniformly on $[0, 1]$.

24.7 Exercise 24.7: Subtracting a power

What we must show. Analyze $f_n(x) = x - x^n$ on $[0, 1]$.

(a) Check the endpoint separately.

For $0 \leq x < 1$, $x^n \to 0$, so the limit is x. At $x = 1$, $f_n(1) = 0$ for every n. Hence

$$f(x) = \begin{cases} x & 0 \leq x < 1, \\ 0 & x = 1. \end{cases}$$

(b) Apply the continuity obstruction.

Each f_n is a polynomial and therefore continuous on $[0,1]$. The limit is discontinuous at 1, because its left-hand limit is 1 and its value there is zero. By Theorem 24.3 the convergence is not uniform.

Conclusion. There is pointwise convergence to the displayed function, but no uniform convergence on $[0,1]$.

24.8 Exercise 24.8: Partial geometric sums at their divergent endpoint

What we must show. Analyze $f_n(x) = \sum_{k=0}^{n} x^k$ on $[0,1]$.

(a) Evaluate the finite sum where its formula is valid.

For $0 \le x < 1$,

$$f_n(x) = \frac{1 - x^{n+1}}{1 - x} \longrightarrow \frac{1}{1 - x}.$$

At $x = 1$ there are $n + 1$ summands equal to 1, so $f_n(1) = n + 1 \to +\infty$. Thus there is *no real-valued pointwise limit on all of* $[0,1]$. An infinite limit at a point is not a real-valued limit function in Definition 24.1.

(b) Uniform convergence would imply pointwise convergence.

Since real-valued pointwise convergence already fails at 1, uniform convergence on $[0,1]$ is impossible.

Conclusion. Convergence occurs pointwise only on $[0,1)$, to $1/(1-x)$; it does not occur pointwise, or uniformly, on the full requested interval.

24.9 Exercise 24.9: A moving peak whose integral vanishes

What we must show. Study $f_n(x) = nx^n(1-x)$ on $[0,1]$.

(a) Fix an input.

At $x = 0$ and $x = 1$ the function is zero. For $0 < x < 1$, the ratio of successive terms of nx^n is $x(1 + 1/n) \to x < 1$, so $nx^n \to 0$. Multiplication by the fixed number $1 - x$ gives $f_n(x) \to 0$. Thus $f = 0$ everywhere on $[0,1]$.

(b) Find a peak that does not shrink to zero.

Differentiating gives

$$f_n'(x) = nx^{n-1}(n - (n+1)x).$$

The sign changes from positive to negative at $x_n = n/(n+1)$, the maximum point. Its value is

$$f_n(x_n) = \frac{n}{n+1}\left(\frac{n}{n+1}\right)^n = \left(\frac{n}{n+1}\right)^{n+1} \longrightarrow e^{-1} > 0.$$

The last limit follows from $(1 + 1/n)^n \to e$. Because the maximum error does not approach zero, convergence is not uniform.

(c) Compute the integrals directly.

Despite the preceding failure,

$$\int_0^1 f_n(x)\,dx = n\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{n}{(n+1)(n+2)} \longrightarrow 0 = \int_0^1 f(x)\,dx.$$

Conclusion. The limit is zero, convergence is not uniform, and the integrals nevertheless converge to the integral of the limit. Uniform convergence is a sufficient condition for this interchange, not a necessary one.

24.10 Exercise 24.10: Adding uniform limits, and the danger with products

What we must show. Prove the sum rule and examine its product analogue.

(a) Split the desired error into two smaller errors.

Let $\varepsilon > 0$. Uniform convergence supplies N_1, N_2 such that $|f_n(x) - f(x)| < \varepsilon/2$ for $n > N_1$ and $|g_n(x) - g(x)| < \varepsilon/2$ for $n > N_2$, in each case for all $x \in S$. For $n > \max\{N_1, N_2\}$ the triangle inequality gives

$$|(f_n + g_n)(x) - (f + g)(x)| \le |f_n(x) - f(x)| + |g_n(x) - g(x)| < \varepsilon.$$

This is uniform convergence of the sums.

(b) Test the product claim before accepting it.

Products need not converge uniformly: on $S = \mathbb{R}$, let $f_n(x) = x$ and $g_n(x) = 1/n$. Their limits are $f(x) = x$ and $g(x) = 0$, both uniform. But $f_n(x)g_n(x) = x/n$ has value 1 at $x = n$, so it fails to converge uniformly to $fg = 0$. A small error multiplied by an unbounded function can become large; this is the missing issue in a naive product proof.

Conclusion. Sums preserve uniform convergence; unrestricted products do not.

24.11 Exercise 24.11: The product counterexample in full

What we must show. Verify every part of the example $f_n(x) = x$, $g_n(x) = 1/n$ on $\mathbb{R}$.

(a) Compute the two errors separately.

With $f(x) = x$ we have $|f_n(x) - f(x)| = 0$ for every n, x. With $g(x) = 0$ we have $|g_n(x) - g(x)| = 1/n$, independently of x. The first error is already zero, and the second tends to zero, so both convergences are uniform.

(b) Compute the product error at a suitable input.

For each n, use $x_n = n$:

$$|f_n(x_n)g_n(x_n) - f(x_n)g(x_n)| = |n/n - 0| = 1.$$

No common index can make all these errors smaller than $1/2$.

Conclusion. $f_n \to f$ and $g_n \to g$ uniformly, but $f_n g_n \not\to fg$ uniformly on $\mathbb{R}$.

24.12 Exercise 24.12: The supremum-error characterization

What we must show. Prove that $f_n \to f$ uniformly on S if and only if $E_n = \sup_{x \in S} |f_n(x) - f(x)| \to 0$. For an empty domain the assertion is vacuous; take its maximum error to be zero. For a nonempty domain, E_n is initially allowed to be $+\infty$.

Uniform convergence implies vanishing suprema.

Given $\varepsilon > 0$, use uniform convergence with $\varepsilon/2$. For sufficiently large n, all the errors are smaller than $\varepsilon/2$, so their supremum satisfies $0 \le E_n \le \varepsilon/2 < \varepsilon$. Notice the non-strict inequality at the supremum: a least upper bound need not be attained. The extra factor $1/2$ avoids any difficulty.

Vanishing suprema imply uniform convergence.

Given $\varepsilon > 0$, choose N with $E_n < \varepsilon$ for all $n > N$. Every individual error is at most its supremum, so $|f_n(x) - f(x)| \le E_n < \varepsilon$ for all $x \in S$ and $n > N$. This is exactly the definition of uniform convergence.

Conclusion. The two conditions are equivalent. The supremum is over the input variable first; the limit is then taken over the index.

24.13 Exercise 24.13: Uniform limits of uniformly continuous functions

What we must show. Show that a uniform limit f of uniformly continuous f_n on (a, b) is uniformly continuous there.

Plan. Approximate f by one fixed f_n, and let that function control nearby inputs. There will be three errors, each allocated $\varepsilon/3$.

Choose one approximating function before choosing the inputs.

Given $\varepsilon > 0$, uniform convergence supplies an index m with $|f_m(t) - f(t)| < \varepsilon/3$ for all $t \in (a, b)$. Uniform continuity of this particular f_m supplies $\delta > 0$ such that $|f_m(x) - f_m(y)| < \varepsilon/3$ whenever $x, y \in (a, b)$ and $|x - y| < \delta$.

Join the three estimates.

For such x, y,

$$|f(x) - f(y)| \le |f(x) - f_m(x)| + |f_m(x) - f_m(y)| + |f_m(y) - f(y)| < \varepsilon.$$

Neither m nor δ was chosen using x or y, so this is uniform continuity, rather than merely continuity at a single point.

Conclusion. The uniform limit is uniformly continuous. The same proof works on any subset of $\mathbb{R}$, not just an interval.

24.14 Exercise 24.14: A narrow peak near zero

What we must show. Analyze $f_n(x) = nx/(1 + n^2x^2)$ and the proposed limit $f = 0$.

(a) Fix x and distinguish zero from nonzero.

At $x = 0$ every term is zero. If $x \neq 0$, dividing by n^2x^2 gives

$$f_n(x) = \frac{1/(nx)}{1 + 1/(n^2x^2)} \longrightarrow 0.$$

Thus pointwise convergence holds on $\mathbb{R}$.

(b) Test the shrinking input $x_n = 1/n$ on $[0, 1]$.

At this input, $f_n(x_n) = 1/2$. Therefore uniform convergence to zero fails on $[0, 1]$. Each fixed positive input eventually lies past the peak, but the peak itself remains of height $1/2$.

(c) Keep a fixed positive distance from zero.

For $x \geq 1$,

$$0 \leq \frac{nx}{1 + n^2x^2} \leq \frac{nx}{n^2x^2} = \frac{1}{nx} \leq \frac{1}{n}.$$

The upper bound tends to zero independently of x, proving uniform convergence on $[1, \infty)$.

Conclusion. Pointwise convergence holds everywhere; uniform convergence fails on $[0, 1]$ and holds on $[1, \infty)$.

24.15 Exercise 24.15: A boundary layer with limiting height one

What we must show. Analyze $f_n(x) = nx/(1 + nx)$ on $[0, \infty)$.

(a) Identify the limit.

At zero the value is zero. If $x > 0$, then $f_n(x) = 1 - 1/(1 + nx) \to 1$. Thus $f(0) = 0$ and $f(x) = 1$ for $x > 0$.

(b) Inspect the interval containing the boundary layer.

The continuous f_n have a discontinuous pointwise limit at zero, so convergence cannot be uniform on $[0, 1]$. Explicitly, $x_n = 1/n$ gives $f_n(x_n) = 1/2$ and $f(x_n) = 1$, an error of $1/2$.

(c) Bound the error on $[1, \infty)$.

For $x \geq 1$ the limit is 1, and

$$|f_n(x) - 1| = \frac{1}{1 + nx} \leq \frac{1}{1 + n} \longrightarrow 0.$$

This is an input-independent bound and hence proves uniform convergence.

Conclusion. The limit jumps from 0 to 1 at zero; convergence is not uniform on $[0, 1]$ but is uniform on $[1, \infty)$.

24.16 Exercise 24.16: A boundary layer with an unbounded limiting function

What we must show. Analyze $f_n(x) = nx/(1 + nx^2)$ for $x \geq 0$.

(a) Compute the pointwise limit.

At zero the value is zero. For $x > 0$, division by n gives $f_n(x) = x/(1/n + x^2) \to 1/x$. Thus $f(0) = 0$ and $f(x) = 1/x$ for $x > 0$.

(b) Show failure of uniform convergence on $[0, 1]$.

One obstruction is the discontinuity, indeed unboundedness, of the limit near zero. More directly, at $x_n = 1/\sqrt{n}$,

$$f_n(x_n) = \frac{\sqrt{n}}{2}, \qquad f(x_n) = \sqrt{n}, \qquad |f_n(x_n) - f(x_n)| = \frac{\sqrt{n}}{2}.$$

These errors grow, so they certainly do not tend uniformly to zero.

(c) Estimate the error away from the singular endpoint.

For $x \geq 1$,

$$\left| \frac{nx}{1 + nx^2} - \frac{1}{x} \right| = \frac{1}{x(1 + nx^2)} \leq \frac{1}{1 + n} \longrightarrow 0.$$

Conclusion. The displayed piecewise f is the limit; convergence fails to be uniform on $[0, 1]$ and is uniform on $[1, \infty)$.

24.17 Exercise 24.17: Passing to the limit while the input also moves

What we must show. If $f_n \to f$ uniformly on $[a, b]$, the f_n are continuous, and $x_n \to x$ in that interval, prove $f_n(x_n) \to f(x)$.

First secure continuity of the limiting function.

By Theorem 24.3, the uniform limit f is continuous on $[a, b]$. Also $x \in [a, b]$, since this interval is closed. Thus continuity gives $f(x_n) \to f(x)$.

Separate the changing function from the changing input.

For every n,

$$|f_n(x_n) - f(x)| \leq |f_n(x_n) - f(x_n)| + |f(x_n) - f(x)|.$$

Given $\varepsilon > 0$, uniform convergence makes the first term smaller than $\varepsilon/2$ for all sufficiently large n, *even at the moving input* x_n. Continuity and $x_n \to x$ make the second term smaller than $\varepsilon/2$ for all sufficiently large n. Taking n beyond both thresholds proves the desired limit.

Conclusion. $f_n(x_n) \to f(x)$. Uniform convergence is what allows us to substitute a different input at every index.

25 More on uniform convergence

Three useful tools. A uniform error bound can be integrated on a finite interval: $|\int_a^b (f_n - f)| \le (b-a)\sup|f_n - f|$. A series of functions converges uniformly precisely when its partial sums do. Finally, the Weierstrass M-test says that $|g_k(x)| \le M_k$ for all inputs, together with $\sum M_k < \infty$, guarantees uniform convergence of $\sum g_k$. If all summands are continuous, its sum is continuous too. An M-test bound must not depend on the input x.

25.1 Exercise 25.1: Bounding an integral by the integral of an absolute value

What we must show. Deduce $|\int_a^b g| \le \int_a^b |g|$ from monotonicity of the integral. The integrals in the statement are assumed to exist.

Place the function between two integrable bounds.

For every x, $-|g(x)| \le g(x) \le |g(x)|$. Applying monotonicity first to the lower pair and then to the upper pair gives

$$-\int_a^b |g(x)|\,dx \le \int_a^b g(x)\,dx \le \int_a^b |g(x)|\,dx.$$

Write the middle integral as A and the nonnegative right-hand integral as B. The statement $-B \le A \le B$ is exactly the statement $|A| \le B$.

Conclusion. $\left|\int_a^b g(x)\,dx\right| \le \int_a^b |g(x)|\,dx.$

25.2 Exercise 25.2: Powers divided by their indices

What we must show. Show uniform convergence of $f_n(x) = x^n/n$ on $[-1,1]$.

Find a common error bound.

Since $|x| \le 1$, $|x^n/n| \le 1/n$ for every input in this interval. Therefore the pointwise limit is $f(x) = 0$. Given $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. Then $n > N$ implies $|f_n(x) - 0| \le 1/n < \varepsilon$ for *all* $x \in [-1,1]$ simultaneously.

Conclusion. The convergence is uniform and the limit function is identically zero.

25.3 Exercise 25.3: A rational trigonometric expression and its integral

What we must show. Study $f_n(x) = (n + \cos x)/(2n + \sin^2 x)$ on $\mathbb{R}$.

(a) Guess the limit, then calculate its exact error.

Dividing by n suggests the limit $1/2$. Subtract it with a common denominator:

$$f_n(x) - \frac{1}{2} = \frac{2\cos x - \sin^2 x}{2(2n + \sin^2 x)}.$$

Since $|2\cos x - \sin^2 x| \le 2 + 1 = 3$ and the denominator is at least $4n$,

$$\left|f_n(x) - \frac{1}{2}\right| \le \frac{3}{4n} \qquad (x \in \mathbb{R}).$$

The right-hand side tends to zero independently of x, proving uniform convergence.

(b) Apply the integral theorem after checking its hypotheses.

The denominator is positive, so every f_n is continuous on $[2, 7]$. The convergence is uniform on that interval by part (a). Theorem 25.2 therefore gives

$$\lim_{n\to\infty} \int_2^7 f_n(x)\,dx = \int_2^7 \frac{1}{2}\,dx = \frac{5}{2}.$$

Conclusion. The uniform limit on $\mathbb{R}$ is $1/2$, and the integral limit is $5/2$.

25.4 Exercise 25.4: Uniform convergence implies the uniform Cauchy property

What we must show. Show that $f_n \to f$ uniformly on S makes (f_n) uniformly Cauchy.

Use the same approximating limit for two indices.

Given $\varepsilon > 0$, choose N so that $|f_k(x) - f(x)| < \varepsilon/2$ whenever $k > N$ and $x \in S$. For *both* $m > N$ and $n > N$, the triangle inequality then gives

$$|f_n(x) - f_m(x)| \le |f_n(x) - f(x)| + |f(x) - f_m(x)| < \varepsilon \qquad (x \in S).$$

The same N works for every pair of sufficiently large indices and every input, which is exactly uniform Cauchyness.

Conclusion. Every uniformly convergent sequence is uniformly Cauchy.

25.5 Exercise 25.5: A uniform limit of bounded functions is bounded

What we must show. If all f_n are bounded on S and $f_n \to f$ uniformly, prove that f is bounded.

Plan. The bounds of all the f_n need not be the same. One suitably chosen approximating function is enough.

Choose one function within distance one of the limit.

Uniform convergence gives an index m such that $|f(x) - f_m(x)| < 1$ for all $x \in S$. Since f_m is bounded, some finite M satisfies $|f_m(x)| \le M$ for every $x \in S$.

Add the approximation error to that fixed bound.

For every input,

$$|f(x)| \le |f(x) - f_m(x)| + |f_m(x)| < 1 + M.$$

This single finite bound establishes boundedness of f.

Conclusion. f is bounded; a bound is $M + 1$ for the chosen approximant.

25.6 Exercise 25.6: Absolute summability gives continuity at the endpoints too

What we must show. Relate $\sum |a_k| < \infty$ to the power series on $[-1, 1]$.

(a) Apply the M-test on the full closed interval.

For every $x \in [-1,1]$, $|a_k x^k| \le |a_k|$. The numerical majorant series $\sum |a_k|$ converges by assumption. The M-test therefore gives uniform convergence of $\sum a_k x^k$ on $[-1,1]$. Every partial sum is a polynomial and hence continuous. Its uniform limit is continuous on $[-1,1]$, including the endpoints with the relative notion of continuity.

(b) Check the coefficient condition in the given example.

For $a_n = 1/n^2$, $n \ge 1$, the numerical series $\sum 1/n^2$ converges by the p-series test, since $2 > 1$. Part (a) therefore applies directly to $\sum_{n=1}^{\infty} x^n/n^2$.

Conclusion. Both the general series and the specified example have continuous sums on the whole closed interval $[-1,1]$.

25.7 Exercise 25.7: A uniformly convergent trigonometric series

What we must show. Prove that $\sum_{n=1}^{\infty} \cos(nx)/n^2$ defines a continuous function on $\mathbb{R}$.

Bound every summand on the unbounded domain.

The elementary estimate $|\cos(nx)| \le 1$ gives $|\cos(nx)/n^2| \le 1/n^2$ for all real x. Since $\sum 1/n^2$ converges, the M-test proves uniform convergence on $\mathbb{R}$.

Transfer continuity from finite sums to the limit.

Each summand is continuous, so every partial sum is continuous. Theorem 24.3 applied to this uniformly convergent sequence of partial sums makes the sum continuous on $\mathbb{R}$.

Conclusion. The series converges uniformly on $\mathbb{R}$ to a continuous function.

25.8 Exercise 25.8: A power series uniform on its entire convergence interval

What we must show. Find the radius and endpoint behavior of $\sum_{n=1}^{\infty} x^n/(n^2 2^n)$.

Compute the radius using coefficient roots.

With $a_n = 1/(n^2 2^n)$,

$$|a_n|^{1/n} = \frac{1}{2n^{2/n}} \longrightarrow \frac{1}{2},$$

since $n^{1/n} \to 1$. Thus the radius is $R = 1/(1/2) = 2$.

Bound the terms at all inputs, including the two endpoints.

For $|x| \le 2$,

$$\left|\frac{x^n}{n^2 2^n}\right| \le \frac{1}{n^2}.$$

The M-test proves uniform convergence on $[-2,2]$, not merely on smaller intervals. The summands are polynomials, so the sum is continuous there.

Conclusion. The radius is 2, and the series converges uniformly to a continuous function on its full interval of convergence $[-2,2]$.

25.9 Exercise 25.9: Geometric series on a smaller and a maximal open interval

What we must show. Compare uniform convergence of $\sum_{n=0}^{\infty} x^n$ on $[-a, a]$ and $(-1, 1)$, where $0 < a < 1$.

(a) Estimate the geometric remainder on the smaller interval.

For $|x| < 1$, the sum is $1/(1-x)$ and the partial sum through N has remainder

$$\frac{1}{1-x} - \sum_{n=0}^{N} x^n = \frac{x^{N+1}}{1-x}.$$

For $|x| \le a$, its absolute value is at most $a^{N+1}/(1-a)$, which tends to zero because $0 < a < 1$. The estimate is uniform in x.

(b) Show that terms do not even tend uniformly to zero on $(-1, 1)$.

If the partial sums converged uniformly, their successive differences x^N would converge uniformly to zero: indeed each difference is bounded by the two uniform errors of the corresponding partial sums. But at $x_N = 2^{-1/N} \in (0, 1)$, we have $x_N^N = 1/2$ for every $N \ge 1$. Thus the necessary uniform vanishing of the terms fails.

Conclusion. Uniform convergence holds on every $[-a, a]$ with $a < 1$, and fails on $(-1, 1)$. Pointwise convergence on the latter interval still holds.

25.10 Exercise 25.10: A non-geometric series compared with a geometric one

What we must show. Analyze $\sum x^n/(1+x^n)$ on $[0, 1)$. The starting index may be taken as $n = 0$; its term is just the constant $1/2$ and does not affect any convergence decision below.

(a) Prove convergence for a fixed input.

For $0 \le x < 1$ and $n \ge 1$,

$$0 \le \frac{x^n}{1+x^n} \le x^n.$$

The majorant $\sum_{n=1}^{\infty} x^n$ converges, so comparison proves convergence of the given nonnegative series at every such fixed x.

(b) Obtain a majorant independent of the input in $[0, a]$.

If $0 \le x \le a < 1$, then $x^n/(1+x^n) \le a^n$. Since $\sum a^n$ converges, the M-test proves uniform convergence on $[0, a]$.

(c) Test the necessary uniform vanishing of the summands.

Take $x_n = 2^{-1/n} \in (0, 1)$. Then $x_n^n = 1/2$ and

$$\frac{x_n^n}{1+x_n^n} = \frac{1/2}{3/2} = \frac{1}{3}.$$

Thus the terms do not tend uniformly to zero on $[0, 1)$. Uniform convergence of the series would force that property by subtracting successive partial sums, as in Exercise 25.9. Hence uniform convergence fails.

Conclusion. There is pointwise convergence on $[0,1)$ and uniform convergence on every $[0,a]$ with $a<1$, but no uniform convergence on $[0,1)$.

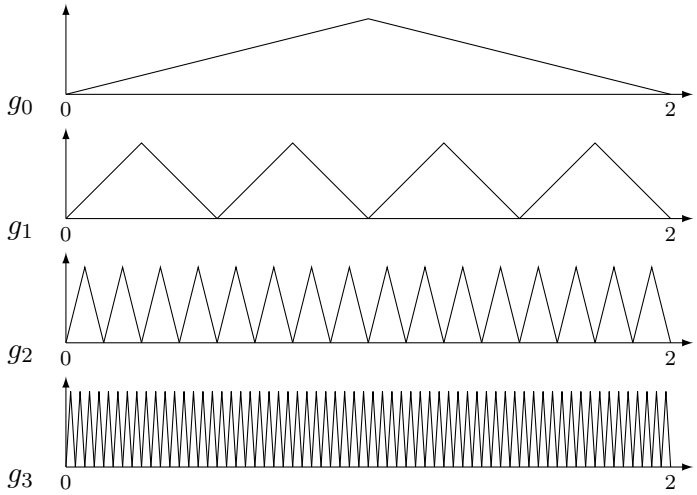
25.11 Exercise 25.11: Four increasingly compressed triangular waves

What we must show. Sketch the $g_n(x) = g(4^n x)$ from Example 3 and prove continuity of $f(x) = \sum_{n=0}^{\infty}(3/4)^n g_n(x)$.

(a) Describe the graphs by exact vertices.

The base function g is the triangular wave of period 2: for every integer j, $g(2j)=0$, $g(2j+1)=1$, and its graph is straight between consecutive integer inputs. Thus g_n has zeros at $2j/4^n$, peaks of height 1 at $(2j+1)/4^n$, and period $2/4^n$. Multiplication of the input compresses the horizontal scale; it does not change the heights.

The sketches below show $0 \le x \le 2$, with the same horizontal and vertical scales in all four rows; repeat each graph periodically in both directions. For g_3, the closely spaced peaks are intentional.



(b) Bound the vertically weighted waves.

The triangular wave is continuous, with $0 \le g \le 1$. Thus every g_n is continuous as the composition of g with the continuous map $x \mapsto 4^n x$, and

$$\left|\left(\frac{3}{4}\right)^n g_n(x)\right| \le \left(\frac{3}{4}\right)^n \qquad (x \in \mathbb{R}).$$

The numerical geometric series $\sum_{n=0}^{\infty}(3/4)^n$ converges. The M-test gives uniform convergence on $\mathbb{R}$, and the uniform-limit theorem gives continuity of f on $\mathbb{R}$.

Conclusion. The graphs have periods 2, 1/2, 1/8, 1/32 and height 1. The weighted infinite sum is continuous everywhere. No differentiability claim is needed for this exercise.

25.12 Exercise 25.12: Integrating a uniformly convergent series

What we must show. If continuous g_k have a uniformly convergent sum g on $[a,b]$, prove $\int_a^b g = \sum_{k=1}^{\infty}\int_a^b g_k$.

Apply the known theorem to partial sums, not directly to infinite sums.

Set $s_n(x) = \sum_{k=1}^{n} g_k(x)$. Each s_n is continuous, and the hypothesis means $s_n \to g$ uniformly. Theorem 25.2 therefore gives $\int_a^b s_n \to \int_a^b g$.

Use linearity only for the finite sum, then take its limit.

For every n,

$$\int_a^b s_n(x)\,dx = \sum_{k=1}^{n} \int_a^b g_k(x)\,dx.$$

Consequently the numerical series on the right converges, and its sum is $\int_a^b g(x)\,dx$ by the definition of an infinite series as a limit of finite partial sums.

Conclusion. Term-by-term integration is valid under the stated uniform convergence and continuity assumptions.

25.13 Exercise 25.13: Adding two uniformly convergent series

What we must show. Show that $\sum(g_k + h_k)$ converges uniformly if $\sum g_k$ and $\sum h_k$ do.

Name the two limits and their partial sums.

Write $G_n = \sum_{k=1}^{n} g_k \to G$ uniformly and $H_n = \sum_{k=1}^{n} h_k \to H$ uniformly. The partial sum of the proposed series is exactly $G_n + H_n$.

Estimate the error of that partial sum.

For every input x,

$$|(G_n + H_n)(x) - (G + H)(x)| \le |G_n(x) - G(x)| + |H_n(x) - H(x)|.$$

Given $\varepsilon > 0$, make each error smaller than $\varepsilon/2$ uniformly, and take the larger of the two index thresholds. This gives uniform convergence to $G + H$.

Conclusion. The sum series is uniformly convergent, with sum equal to the sum of the two original sum functions.

25.14 Exercise 25.14: Multiplying a series by one bounded function

What we must show. If $\sum g_k$ converges uniformly on S and h is bounded there, prove uniform convergence of $\sum h g_k$.

Write the new partial sums as a product with the old ones.

Let $G_n = \sum_{k=1}^{n} g_k \to G$ uniformly and choose a finite $M \ge 0$ with $|h(x)| \le M$ for all $x \in S$. Then

$$\sum_{k=1}^{n} h(x)g_k(x) = h(x)G_n(x), \qquad |h(x)G_n(x) - h(x)G(x)| \le M|G_n(x) - G(x)|.$$

Choose the uniform error tolerance after accounting for the bound.

If $M = 0$, every new term is zero and there is nothing to prove. If $M > 0$, choose N so that $|G_n(x) - G(x)| < \varepsilon/M$ for every $x \in S$ and $n > N$. The displayed inequality then bounds the new error by ε, uniformly in x.

Conclusion. The new series converges uniformly to hG. No continuity of h is required, only its boundedness.

25.15 Exercise 25.15: Monotone convergence on a compact interval: Dini's theorem

What we must show. Prove both monotone versions of the uniform-convergence assertion.

(a) First note that decreasing terms converging to zero are nonnegative.

For each fixed x, $f_n(x) \geq \lim_m f_m(x) = 0$. We must show that the nonnegative errors $f_n(x)$ become small at one common index.

Use continuity to turn each strict error bound into an open set.

Fix $\varepsilon > 0$, and let

$$U_n = \{x \in [a, b] : f_n(x) < \varepsilon\}.$$

Continuity makes U_n relatively open in $[a, b]$. Since $f_{n+1} \leq f_n$, these sets increase: $U_n \subseteq U_{n+1}$. Pointwise convergence to zero implies every x lies in some U_n, so $\{U_n : n \geq 1\}$ is an open cover of $[a, b]$.

Compactness turns the varying pointwise indices into one common index.

The closed bounded interval is compact, so finitely many of the U_n cover it. Let N be the largest of their indices. Because the sets are nested, U_N alone covers the whole interval. Thus $0 \leq f_n(x) \leq f_N(x) < \varepsilon$ for all $x \in [a, b]$ and $n \geq N$. This is uniform convergence to zero.

(b) Reduce increasing convergence to the decreasing case.

Suppose now $f_n(x)$ increases to a continuous function $f(x)$. Set $h_n = f - f_n$. Each h_n is continuous because both f and f_n are. At every input h_n decreases to zero, and $h_n \geq 0$. Part (a) gives $h_n \to 0$ uniformly, which is exactly $f_n \to f$ uniformly.

Conclusion. Both claims hold. Compactness and continuity of the limiting function are essential ingredients in turning the pointwise monotonicity into a uniform bound.

26 Differentiation and integration of power series

The permission comes before the calculation. Inside the radius of convergence of a power series, we may differentiate term by term, and we may integrate term by term from zero to any interior point. Uniform convergence on each smaller closed interval is the reason integration works. Endpoints require a separate argument: convergence inside an interval does not by itself settle either endpoint. We use Abel's theorem only after checking that the numerical endpoint series converges.

26.1 Exercise 26.1: Term-by-term integration at a positive input

What we must show. If $f(t) = \sum_{k=0}^{\infty} a_k t^k$ has radius $R > 0$, prove $\int_0^x f(t)\,dt = \sum_{k=0}^{\infty} a_k x^{k+1}/(k+1)$ when $0 < x < R$.

Work on an interval lying strictly inside the radius.

Fix such an x, and put $s_n(t) = \sum_{k=0}^{n} a_k t^k$. The power series converges uniformly on $[-x, x]$ by Theorem 26.1, and therefore also on $[0, x]$. Each s_n is a continuous polynomial.

Pass the limit through the integral using the established theorem.

Theorem 25.2 yields

$$\int_0^x f(t)\,dt = \lim_{n\to\infty} \int_0^x s_n(t)\,dt.$$

For a finite sum, linearity and the power rule give

$$\int_0^x s_n(t)\,dt = \sum_{k=0}^{n} a_k \int_0^x t^k\,dt = \sum_{k=0}^{n} \frac{a_k}{k+1} x^{k+1}.$$

Taking the limit of these finite sums gives precisely the claimed infinite series.

Conclusion. The term-by-term integration formula holds for every positive $x < R$. The strict inequality $x < R$ is what secured the uniform convergence used in the proof.

26.2 Exercise 26.2: A differentiated geometric series and three numerical sums

What we must show. Derive $\sum_{n=1}^{\infty} nx^n$ and use it to evaluate the requested sums.

(a) Differentiate first, then restore the missing power of x.

For $|x| < 1$, the geometric series is $\sum_{n=0}^{\infty} x^n = 1/(1-x)$. Its radius is 1, so Theorem 26.5 allows differentiation at every such x:

$$\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

Multiplying by x gives

$$\boxed{\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}} \qquad (|x| < 1).$$

(b) Substitute $x = 1/2$.

The value lies strictly inside the radius, so the formula gives

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1/2}{(1-1/2)^2} = \frac{1/2}{1/4} = 2.$$

(c) Substitute the two signed inputs separately.

At $x = 1/3$,

$$\sum_{n=1}^{\infty} \frac{n}{3^n} = \frac{1/3}{(2/3)^2} = \frac{3}{4}.$$

At $x = -1/3$,

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{3^n} = \frac{-1/3}{(1+1/3)^2} = \frac{-1/3}{16/9} = -\frac{3}{16}.$$

Conclusion. The three numerical sums are 2, 3/4, and $-3/16$.

26.3 Exercise 26.3: One more derivative gives squared indices

What we must show. Find $\sum_{n=1}^{\infty} n^2 x^n$ and evaluate it at 1/2 and 1/3.

(a) Differentiate the identity from Exercise 26.2.

For $|x| < 1$, term-by-term differentiation is legitimate, because differentiation preserves the radius of a power series. Hence

$$\sum_{n=1}^{\infty} n^2 x^{n-1} = \frac{d}{dx}\left[\frac{x}{(1-x)^2}\right] = \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^3} = \frac{1+x}{(1-x)^3}.$$

Multiplication by x gives

$$\boxed{\sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}} \qquad (|x| < 1).$$

(b) Evaluate the two substitutions.

Both inputs are strictly inside the radius. At $x = 1/2$ the numerator is $(1/2)(3/2) = 3/4$ and the denominator is $(1/2)^3 = 1/8$, so

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n} = \frac{3/4}{1/8} = 6.$$

At $x = 1/3$ the numerator is $4/9$ and the denominator is $8/27$, so

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} = \frac{4/9}{8/27} = \frac{3}{2}.$$

Conclusion. The general formula is $x(1+x)/(1-x)^3$, and the requested values are 6 and 3/2.

26.4 Exercise 26.4: Integrating the Gaussian power series

What we must show. Express e^{-x^2} and $F(x) = \int_0^x e^{-t^2}\,dt$ as power series.

(a) Substitute into the exponential series.

The exponential identity, with its constant term included, is

$$e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!} \qquad (u \in \mathbb{R}).$$

Taking $u = -x^2$ gives

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots .$$

This substitution is valid for every real x because the original exponential series converges for every real input.

(b) Verify the radius before integrating.

For fixed x, the ratio of successive absolute terms is $x^2/(n+1) \to 0$, so this power series has infinite radius. Theorem 26.4 therefore applies between zero and every real x, whether positive or negative. It gives

$$\boxed{F(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)n!}} = x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots .$$

The lower limit contributes zero for each monomial, so no integration constant has been omitted.

Conclusion. The two displayed expansions hold for all $x \in \mathbb{R}$.

26.5 Exercise 26.5: A function equal to its own derivative, without naming it

What we must show. For $f(x) = \sum_{n=0}^{\infty} x^n/n!$, prove $f' = f$ directly from the series.

Check that term-by-term differentiation is allowed everywhere.

For each fixed x, the ratio of successive absolute terms is $|x|/(n+1) \to 0$. Thus the radius is infinite, and Theorem 26.5 allows term-by-term differentiation at every real input.

Differentiate and carefully change the index.

The constant term differentiates to zero, and for $n \geq 1$, $n/n! = 1/(n-1)!$. Therefore

$$f'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{j=0}^{\infty} \frac{x^j}{j!} = f(x).$$

In the last step we set $j = n - 1$, so the first new index is zero.

Conclusion. $f' = f$ on $\mathbb{R}$. No identification of f with the exponential function was used.

26.6 Exercise 26.6: A trigonometric identity derived without trigonometry

What we must show. For

$$s(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \qquad c(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

prove the requested derivative relations and $s^2 + c^2 = 1$.

(a) Differentiate each series with its radius justified.

For fixed x, ratios of successive absolute terms tend to zero in both series, since their denominators acquire two factorial factors while their numerators acquire only a factor x^2. Thus both radii are infinite. Theorem 26.5 gives

$$s'(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = c(x).$$

For c, the $n = 0$ term differentiates to zero, so

$$\begin{aligned} c'(x) &= \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{(2n-1)!} \\ &= -\sum_{j=0}^{\infty} \frac{(-1)^j x^{2j+1}}{(2j+1)!} = -s(x). \end{aligned}$$

Here $j = n - 1$, and $(-1)^{j+1} = -(-1)^j$ explains the minus sign.

(b) Differentiate the sum of squares.

The product rule gives

$$(s^2 + c^2)' = 2ss' + 2cc' = 2sc + 2c(-s) = 0.$$

(c) Determine the constant from one input.

A differentiable function with derivative zero on an interval is constant there, by the Mean Value theorem. Apply this on the interval joining zero to any real x. Thus $s(x)^2 + c(x)^2 = s(0)^2 + c(0)^2$. The defining series give $s(0) = 0$ and $c(0) = 1$, so that constant is 1.

Conclusion. $s' = c$, $c' = -s$, and $s(x)^2 + c(x)^2 = 1$ for every real x, proved using the series alone.

26.7 Exercise 26.7: Why absolute value has no power-series representation at zero

What we must show. Decide whether $|x| = \sum_{n=0}^{\infty} a_n x^n$ can hold for all real x.

A power series with positive radius is differentiable at its center.

If such a representation held for every real input, its radius would be infinite, in particular positive. Theorem 26.5 would make its sum differentiable at zero.

Calculate the two difference quotients of absolute value.

Since $|0| = 0$,

$$\frac{|h| - |0|}{h} = \begin{cases} 1 & h > 0, \\ -1 & h < 0. \end{cases}$$

The right-hand and left-hand limits differ, so $|x|$ is not differentiable at zero. This contradicts the consequence of a power-series representation.

Conclusion. No such power series exists, even on any two-sided neighborhood of zero. On the separate half-lines the formulas are $|x| = x$ and $|x| = -x$, but these do not combine into one power series about zero. Polynomial approximation, discussed next, is a different and weaker requirement.

26.8 Exercise 26.8: The arctangent series and its endpoints

What we must show. Derive the arctangent series inside $(-1, 1)$ and settle both endpoints.

(a) Substitute into the geometric series.

For $|x| < 1$, put $y = -x^2$, so $|y| < 1$. Then

$$\sum_{n=0}^{\infty} (-1)^n x^{2n} = \sum_{n=0}^{\infty} y^n = \frac{1}{1-y} = \frac{1}{1+x^2}.$$

(b) Integrate on a strictly interior interval.

The power series has radius 1, so Theorem 26.4 gives, for $|x| < 1$,

$$\int_0^x \frac{dt}{1+t^2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}.$$

The integral on the left is $\arctan x - \arctan 0 = \arctan x$. Therefore

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \qquad (-1 < x < 1).$$

(c) Establish convergence at 1 before using Abel's theorem.

At $x = 1$ the numerical series is $\sum_{n=0}^{\infty} (-1)^n/(2n+1)$. Its positive term sizes decrease to zero, so the alternating-series test proves convergence. Abel's theorem then identifies its sum with the limit of the interior sum as $x \uparrow 1$. Since arctangent is continuous,

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \lim_{x \uparrow 1} \arctan x = \frac{\pi}{4}.$$

Multiplication by 4 gives the familiar formula

$$\boxed{\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)}.$$

(d) Evaluate at the negative endpoint with its sign intact.

Every power in the series is odd. Substitution $x = -1$ therefore makes the series the negative of the convergent series in part (c). Its sum is $-\pi/4 = \arctan(-1)$. It converges conditionally at both endpoints: the series of absolute values $\sum 1/(2n+1)$ diverges by comparison with the harmonic series, since $1/(2n+1) \geq 1/(2n+2)$.

Conclusion. The arctangent expansion holds on $[-1, 1]$, with endpoint values $\pi/4$ and $-\pi/4$ justified separately, rather than by interior convergence alone.

27 Weierstrass's approximation theorem

Approximation is different from equality with a power series. A continuous function on a closed bounded interval can be approximated uniformly by polynomials, even when it has corners. The approximating polynomials need not be successive partial sums of a single power series. The Bernstein polynomials on $[0,1]$ are

$$B_n f(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k}x^k(1-x)^{n-k} \qquad (n \geq 1).$$

For $x \in [0,1]$, the weights $\binom{n}{k}x^k(1-x)^{n-k}$ are nonnegative and sum to 1. Thus $B_n f(x)$ is a weighted average of sampled values of f. Products with exponent zero are interpreted as 1, including at endpoints.

27.1 Exercise 27.1: Transferring approximation from $[0,1]$ to $[a,b]$

What we must show. Deduce the general Weierstrass approximation theorem from the Bernstein theorem on $[0,1]$.

Move the input interval to the standard one.

Assume first $a < b$. The affine function $\phi(t) = a + (b-a)t$ maps $[0,1]$ onto $[a,b]$, with inverse $\phi^{-1}(x) = (x-a)/(b-a)$. If f is continuous on $[a,b]$, then $g = f \circ \phi$ is continuous on $[0,1]$. Theorem 27.4 supplies polynomials $q_n = B_n g$ converging uniformly to g there.

Move the approximating polynomials back.

Define

$$p_n(x) = q_n\left(\frac{x-a}{b-a}\right).$$

A polynomial composed with an affine function is again a polynomial. For every $x \in [a,b]$, let $t = (x-a)/(b-a) \in [0,1]$. Then

$$|p_n(x) - f(x)| = |q_n(t) - g(t)|.$$

Because the substitution is a bijection between the intervals, the suprema of these errors on the two intervals are equal. The supremum on $[0,1]$ tends to zero, so the one on $[a,b]$ does too.

Include the degenerate interval if needed.

If $a = b$, the constant polynomial $p_n(x) = f(a)$ is already exact on the one-point interval.

Conclusion. Every continuous function on $[a,b]$ is a uniform limit of polynomials.

27.2 Exercise 27.2: One sequence that works on every bounded subset of $\mathbb{R}$

What we must show. For continuous f on $\mathbb{R}$, find polynomials p_n converging uniformly to f on each bounded subset.

Use a larger interval and a smaller tolerance at each stage.

For each integer $n \geq 1$, Weierstrass approximation on $[-n, n]$ gives one polynomial p_n with

$$|p_n(x) - f(x)| < \frac{1}{n} \qquad (|x| \leq n).$$

We choose one such polynomial for each n.

Verify convergence on an arbitrary bounded set.

Let $E \subseteq \mathbb{R}$ be bounded. Choose M so that $|x| \leq M$ for every $x \in E$. Given $\varepsilon > 0$, choose an integer N larger than both M and $1/\varepsilon$. For $n > N$, every $x \in E$ belongs to $[-n, n]$, so $|p_n(x) - f(x)| < 1/n < \varepsilon$.

Conclusion. The constructed sequence converges uniformly on every bounded subset of $\mathbb{R}$. The index threshold can depend on that bounded set; this does not assert uniform convergence on all of $\mathbb{R}$.

27.3 Exercise 27.3: Why the entire real line is too large

What we must show. Show that neither $\sin x$ nor e^x is a uniform limit of polynomials on $\mathbb{R}$.

(a) A polynomial uniformly close to sine would have to be constant.

Suppose $p_n \to \sin$ uniformly. Then for some index m, $|p_m(x) - \sin x| < 1/2$ for every real x. Consequently $|p_m(x)| \leq |\sin x| + 1/2 \leq 3/2$ everywhere. A nonconstant polynomial is unbounded on $\mathbb{R}$: its leading term dominates all terms of smaller degree as $|x| \to \infty$. Thus p_m must be a constant c. At $x = \pi/2$ and $x = 3\pi/2$, the proposed approximation requires $|c - 1| < 1/2$ and $|c + 1| < 1/2$. But the triangle inequality would then give $2 \leq |1 - c| + |c + 1| < 1$, an impossibility.

(b) No polynomial can stay within a fixed error of the exponential.

Let p be any polynomial. Choose an integer $d \geq 0$ at least its degree, and $C > 0$ such that $|p(x)| \leq Cx^d$ for $x \geq 1$; the sum of the absolute values of its coefficients, enlarged if necessary, supplies such a C. The exponential series has nonnegative terms at nonnegative inputs, so

$$e^x \geq \frac{x^{d+1}}{(d+1)!} \qquad (x \geq 0).$$

Therefore, for $x \geq 1$,

$$e^x - p(x) \geq \frac{x^{d+1}}{(d+1)!} - Cx^d = x^d \left(\frac{x}{(d+1)!} - C \right) \longrightarrow +\infty.$$

In particular, no polynomial has $|p(x) - e^x| < 1$ for all real x. Uniform convergence would eventually require exactly such a polynomial, so it is impossible.

Conclusion. Uniform polynomial approximation on each bounded interval does not extend to uniform approximation on the whole real line in either example.

27.4 Exercise 27.4: Correcting the two endpoint values

What we must show. Construct approximating polynomials with $p_n(a) = f(a)$ and $p_n(b) = f(b)$.

Start with any uniformly approximating sequence.

For $a < b$, choose polynomials $q_n \to f$ uniformly on $[a, b]$. Write $E_n = \sup_{x\in[a,b]} |q_n(x) - f(x)|$, so $E_n \to 0$. The desired corrections at the two endpoints are $A_n = f(a) - q_n(a)$ and $B_n = f(b) - q_n(b)$.

Interpolate those corrections with a straight line.

Define

$$s_n(x) = \frac{b-x}{b-a}A_n + \frac{x-a}{b-a}B_n, \qquad p_n(x) = q_n(x) + s_n(x).$$

The function s_n is affine, so p_n is a polynomial. At $x = a$ its correction is A_n, and at $x = b$ its correction is B_n. Thus $p_n(a) = f(a)$ and $p_n(b) = f(b)$ exactly.

Show that the correction stays small throughout the interval.

On $[a, b]$ the two coefficients in s_n are nonnegative and sum to 1. Also $|A_n|, |B_n| \le E_n$. Hence $|s_n(x)| \le E_n$ for all such x, and

$$|p_n(x) - f(x)| \le |q_n(x) - f(x)| + |s_n(x)| \le 2E_n \longrightarrow 0.$$

The bound is uniform in x. If $a = b$, the constant polynomial $f(a)$ settles the one-point case directly.

Conclusion. The endpoint-corrected polynomials retain uniform convergence.

27.5 Exercise 27.5: Bernstein polynomials for the first two powers

What we must show. Compute $B_n f$ when $f(x) = x$ and when $f(x) = x^2$.

(a) Use the first binomial moment.

For $f(t) = t$,

$$B_n f(x) = \frac{1}{n}\sum_{k=0}^{n} k\binom{n}{k}x^k(1-x)^{n-k}.$$

For $k \ge 1$, $k\binom{n}{k} = n\binom{n-1}{k-1}$. Substitution and the binomial theorem give

$$\sum_{k=0}^{n} k\binom{n}{k}x^k(1-x)^{n-k} = nx\sum_{j=0}^{n-1}\binom{n-1}{j}x^j(1-x)^{n-1-j} = nx.$$

Therefore $B_n f(x) = x$ for every $n \ge 1$: linear functions of this form are reproduced exactly.

(b) Decompose the square into two binomial moments.

For $f(t) = t^2$, use $k^2 = k(k-1) + k$. For $n \ge 2$, the factorial identity $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$ gives

$$\sum_{k=0}^{n} k(k-1)\binom{n}{k}x^k(1-x)^{n-k} = n(n-1)x^2.$$

The same identity is zero on both sides for $n = 1$. Adding the first moment nx and dividing by n^2 yields

$$B_n f(x) = \frac{n(n-1)x^2 + nx}{n^2} = x^2 + \frac{x(1-x)}{n}.$$

As a check, $0 \le x(1-x) \le 1/4$ on $[0, 1]$, so the error is at most $1/(4n)$, consistent with uniform convergence.

Conclusion. $B_n(t \mapsto t)(x) = x$ and $B_n(t \mapsto t^2)(x) = x^2 + x(1-x)/n$.

27.6 Exercise 27.6: A necessary condition for uniform Bernstein approximation

What we must show. If $B_n f \to f$ uniformly on $[0,1]$, prove that f is continuous, without initially assuming continuity of f.

Notice what kind of function each approximant is.

For each fixed n, the finitely many sampled values $f(k/n)$ are real constants. Thus the formula defining $B_n f$ is a finite linear combination of polynomials, so $B_n f$ is a polynomial and is continuous on $[0,1]$.

Use the uniform-limit theorem.

The assumption is uniform convergence of this sequence of continuous functions to f. Theorem 24.3 therefore makes f continuous on $[0,1]$.

Conclusion. Continuity is necessary for uniform convergence of the Bernstein polynomials to the original function. Together with Theorem 27.4, it is also sufficient.

27.7 Exercise 27.7: A Bernstein bound from nonnegative weights

What we must show. If $|f(t)| \le M$ for every $t \in [0,1]$, prove $|B_n f(x)| \le M$ for all $n \ge 1$ and $x \in [0,1]$.

Use the triangle inequality with signs of the weights checked.

For $0 \le x \le 1$, every weight $w_k = \binom{n}{k} x^k (1-x)^{n-k}$ is nonnegative. Also each sample point k/n belongs to $[0,1]$, so $|f(k/n)| \le M$. Hence

$$|B_n f(x)| \le \sum_{k=0}^{n} |f(k/n)| w_k \le M \sum_{k=0}^{n} w_k.$$

Evaluate the total weight.

By the binomial theorem, $\sum_{k=0}^{n} w_k = (x + (1-x))^n = 1$. The preceding bound is therefore M.

Conclusion. Every Bernstein polynomial obeys the same bound M as f on $[0,1]$. The interval matters: outside it the weights need not be nonnegative.